



Quasicrystals

Particles with curious atomic arrangements that are superficially similar to crystals. <http://digit.in/mar21-21>



Wasted heat as electricity

Silver Antimony Telluride is a material that converts wasted heat from home appliances into electricity. <http://digit.in/mar21-22>

Remixing Materials

Creating composite materials with desirable properties

Aditya Madanapalle | aditya@digit.in

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esearchers developing new materials for industrial, electronic and biomedical applications blend

various components together to create better chips, wafers, coatings and separation membranes. These are used in everything from artificial lungs to semiconductors. For solar cells and light emitting diodes, the improvements in performance has been due to optimising the blends between organic and inorganic components. Arriving at the right blend of materials is however a slow, painstaking and inconsistent process. The properties of the material depends on the fine grain structure and its constituent elements. The blended

material has to have the right mix of properties derived from its constituent atoms and molecules, as well as more complex structures such as polymers and nanoparticles. The properties of the blended material originate from the nanoscale structures, features and patterns formed as a consequence of the blending process. The conventional approach for creating blended materials so far has been to create samples of materials with different proportions of blended constituents. The property of this blended material sample is studied, then another sample is created with a different proportion of blended constituents. Once a large enough library of samples has been built up, the material can be characterised and the optimum blend identified.

SYNTHESIS AND CHARACTERISATION OF BLENDS

A team of researchers from Brookhaven National Laboratory, Yale University, and the University of Pennsylvania have come up with a tool that promises to accelerate the process. There

are two parts of the problem here, the first is synthesising the new materials, and the second is characterising it and probing its properties. The first half of the problem was solved by building a tool that consists of three component solutions in syringe pumps that spray electrically charged particles at a heated substrate. The substrate changes position below the pumps, and the rate of flow of the solutions in each of the pumps can be controlled, allowing researchers to program a recipe. The really clever part is that the components can be deposited as gradients on the substrate, allowing researchers to synthesise a single sample, that actually acts as a library for a large number of different blended materials. Within the single component, up to three materials can be blended in smooth gradients. Existing combinatorial methods for exploring compositions mostly only explore properties of two blended components.

Once the sample is synthesised and the blended material is created using the electrospray deposition tool, it has to be characterised. To